Intravascular Micro-Sieve System for Localized Drug Delivery and Target Particle Capture (IMS-LDD-TPC)

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Introduction: In modern medical science, targeted drug delivery remains a significant challenge in cancer treatment. Conventional systemic drug administration distributes therapeutic agents throughout the body, potentially damaging healthy tissues and causing undesirable side effects. This motivates us to develop a localized drug delivery approach that will concentrate therapeutic agents near the target area and minimize the exposure to surrounding healthy tissues.

Proposed System and Methodology: The proposed *Intravascular Micro-Sieve System for Localized Drug Delivery and Target Particle Capture (IMS-LDD-TPC)* integrates localized drug delivery and target-particle capture within a single retrievable intravascular device. Its central component, the *Intravascular Micro-Cage (IMC)*, is a cylindrical device having fine-mesh sieve plates on the base and the top, and houses a drug-carrier capsule. An external acoustic excitation is proposed to induce capsule vibration and controlled release of a model therapeutic agent (e.g. colored water) near the target region. The micro-sieve is intended to retain suitably sized particles (target particles) while permitting continuous fluid passage through the device.

Proposed Block Diagram:

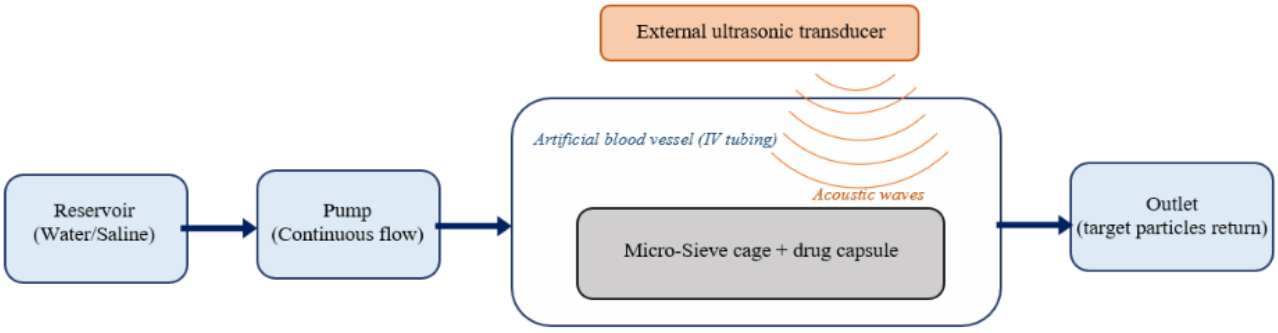


Fig 1: Proposed Block Diagram

Proposed Outcome: The project aims to develop a functional laboratory-scale prototype demonstrating the engineering feasibility of integrating controlled localized model-drug release and selective physical retention of larger target particles within a single compact, retrievable intravascular device while allowing continuous fluid passage. Experimental evaluation will investigate acoustic actuation, release characteristics, particle-separation performance, and device retrieval. The findings are expected to identify practical design limitations and establish a foundation for further investigation and optimization of the proposed system.

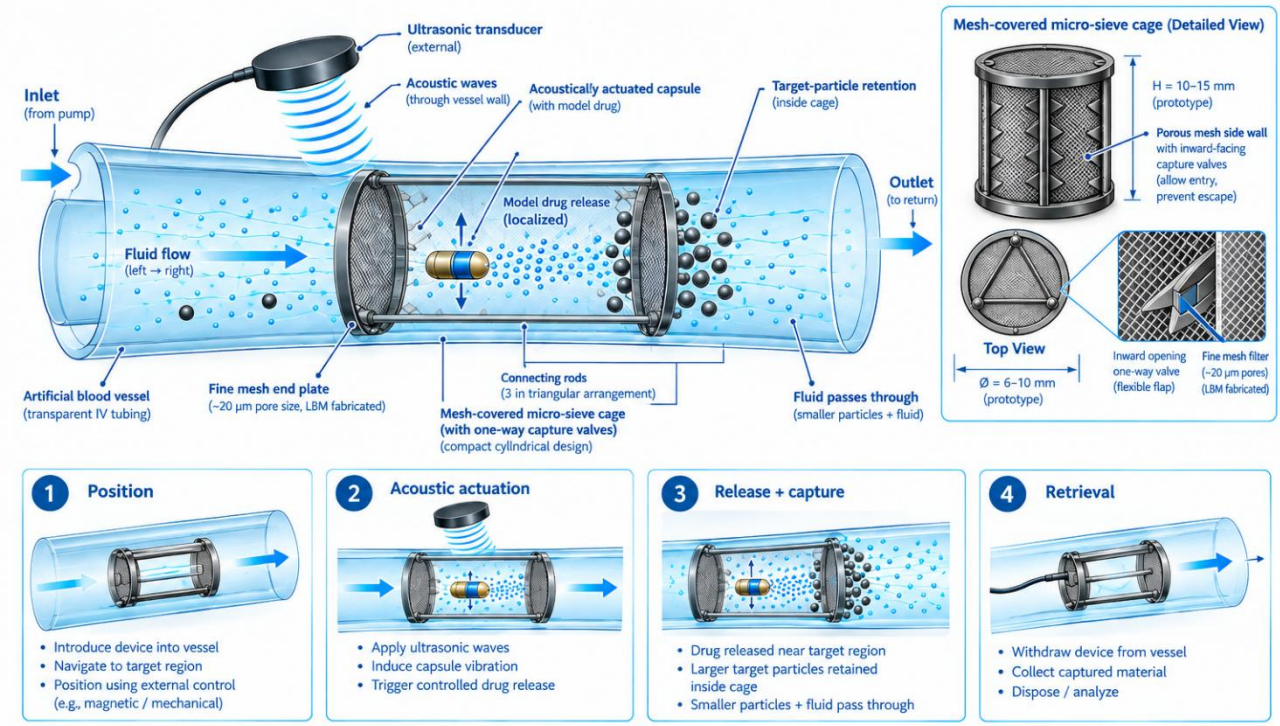


Fig 2: Proposed IMS-LDD-TPC System: Design and Working Principle